ZZ-M-81C
October 13, 1969
SUPERSEDING
Int. Fed. Spec. ZZ-M-0081B(GSA-FSS)
June 26, 1967 and
Fed. Spec. ZZ-M-81A
August 7, 1952

FEDERAL SPECIFICATION

MATTING, RUBBER AND PLASTIC (SPECIAL, DIELECTRIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers special dielectric rubber and plastic matting capable of providing protection from electric grounds with a potential not exceeding 3000 volts.

1.2 Classification.

1.2.1 Classes. Matting shall be furnished in the following classes as specified (see 6.2):

Class 1 - Rubber, fire resistant.
Class 2 - Rubber, standard.
Class 3 - Plastic, standard.
Class 4 - Vinyl, fire resistant.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

UU-P-268 - Paper, Kraft, Untreated, Wrapping.
MMM-A-260 - Adhesive, Water-Resistant (For Sealing Waterproofed Paper).
PPP-B-585 - Boxes, Wood, Wirebound.

FSC 7220

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PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055 - Barrier Material, Waterproof Flexible.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper Water Resistant (For Carton Sealing).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 501 - Floor Coverings, Resilient, Non-Textile; Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or proposal shall apply.

Nongovernmental:

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Class 1 and class 2 matting. The rubber matting shall be made of synthetic rubber or a mixture of synthetic and reclaimed rubber having a fabric backing or insert. The rubber compound shall be properly vulcanized, be free of fiber, and have no offensive odor. The matting shall meet the requirements specified in table I.

TABLE I. Physical properties of matting

Characteristics	Class 1	Class 2	Classes 3 & 4
Weight of fabric, minimum oz/sq.yd.	4 3/4	4 3/4	-
Tensile strength, minimum p.s.i.	1200	700	1600
Maximum decrease after			
Immersion in sulfuric acid, percent of original	30	30	30
Exposure to light, percent of original	35	35	35
Air oven aging, percent of original	20	20	20
Ultimate elongation of compound			
Original, minimum, percent	250	250	150
Maximum decrease after immersion in sulfuric acid, percent of original	25	25	25
Exposure to light, percent of original	25	25	25
Air oven aging, percent of original	25	25	25
Set of compound, maximum, percent	35	35	15
Friction (adhesion of fabric to rubber compound) minimum lb/in. of width	4	4	-

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3.1.1 Fabric. The fabric shall be cotton sheeting evenly and firmly woven from good quality cotton.

3.2 Class 3 and class 4 matting. The plastic matting shall be a suitable, compounded, virgin polymer or copolymer of vinyl chloride, vinylidene chloride or vinyl acetate resins combined with suitable plasticizer, stabilizer, fillers and pigments. The compound shall be uniform, free of fiber and have no objectionable odor. Virgin polymer shall be defined as a polymer or copolymer that has not been processed into a finished product prior to use in the specified matting. The matting shall meet the requirements specified in table I.

3.3 Construction.

3.3.1 Class 1 and class 2. Rubber matting shall be fabricated as a layer of rubber backed with one ply of cotton sheeting or layers of rubber with cotton sheeting between layers. The back surface of the matting shall be smooth or have a fabric imprint. The wearing surface shall be corrugated or have designs as specified in 3.3.3.

3.3.2 Class 3 and class 4. Plastic matting shall be of commercial design with longitudinal, diamond, pyramid, knob-shaped, or simulated fabric-weave corrugations. The back of the matting shall be finished with a knurled surface.

3.3.3 Wearing surface. Unless otherwise specified (see 6.2), the wearing surface of class 1 and class 2 matting may have any of the following designs:

a. Corrugations formed by triangular grooves approximately 1/8-inch apart and approximately 1/16-inch deep, or flat-bottomed grooves approximately 3/32-inch wide at the bottom and 3/64-inch deep alternating with flat-topped summits approximately 3/32-inch wide.

b. Hemispherical knobs approximately 1/4 inch in diameter spaced approximately 1/2 inch apart, or diamond-shaped raised figures. The depth of the design shall be approximately 1/16 inch

3.4 Dimensions.

3.4.1 Length and width. The length and width of the roll shall be as specified (see 6.2 and 6.3).

3.4.1.1 Tolerances. Tolerances of the matting shall be as follows:

Width - 36 inches and under ± 1/2 inch.
Over 36 inches ± 3/4 inch.
Length - Roll of 25 linear yards, plus 12 inches, with no more than 2 pieces per roll.

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3.4.2 Thickness. Class 1 and class 2 shall have a thickness of 1/4 inch ± 1/32 inch. The thickness of class 3 and class 4 matting shall be 3/16 inch plus 1/32, minus 0.

3.5 Color. The color of the matting shall be as specified (see 6.2). Normally, class 1 and class 2 matting shall be furnished in black.

3.6 Physical requirements.

3.6.1 Volatile matter (class 3 and class 4). The amount of volatile matter in the matting shall not exceed 1.0 percent.

3.6.2 Flexibility.

3.6.2.1 At temperature of 23° ± 5° C. The matting shall be subjected to light aging and then tested for volatility. After the volatility test, the matting shall not crack or show any indication of weakness and shall retain its flexibility.

3.6.2.2 At temperature of 0° ± 1° C. The matting shall not crack or break when tested for flexibility at this temperature.

3.6.3 Fire resistance, class 1 and class 4. Class 1 and class 4 matting shall exhibit no more than a slight flame or flash during the test. The area of the burned portion may be slightly greater than the area exposed to direct heat, but shall not extend more than 10 inches from the firing end of the test plate.

3.6.4 Electrical properties.

3.6.4.1 Voltage. The entire area of the matting shall withstand 15,000 volts for one minute without puncturing or becoming appreciably warm at any spot, or showing other evidence of weakness.

3.6.4.2 Dielectric strength. The matting shall not fail when subjected to a charge not less than 40,000 volts.

3.7 Identification marking. The matting shall be permanently and legibly marked on the backside, in not less than two foot-increments, with the name of the manufacturer or his registered trademark, specification number, and the date of curing.

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3.8 Workmanship. The finished matting shall be free from embedded foreign material, cracks, tears, or objectionable odor and shall conform to the quality assurance provisions and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Unless otherwise specified (see 6.2), the supplier shall furnish a certificate with each shipment or lot stating that materials and components are in compliance with the requirements specified. The Government reserves the right to test the items of the shipment or lot to determine the validity of the certification.

4.2 Sampling. The sample size for levels of inspection and the acceptable quality levels shall be in accordance with MIL-STD-105.

4.2.1 Lot. A lot shall consist of all matting of the same class and size submitted for inspection at the same time.

4.2.2 Sampling for dimensional and visual examination of end item. Sampling of matting for dimensional and visual examination shall be in accordance with inspection level II.

4.2.3 Sampling for test. Sampling for tests shall be in accordance with inspection level S-2.

4.3 Visual and dimensional examination of end item. The matting selected in accordance with 4.2.2 shall be examined for the defects listed in table II. The AQL's shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects. A sample unit for this examination shall consist of a completely fabricated roll of matting.

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TABLE II. Classification of defects

Examine	Defects	Major	Minor
End item	Material not fabricated as specified.	X	
	Not color specified.	X	
	Has an objectionable odor.	X	
	Width and thickness not as specified.	X	
	Length not as specified.		X
	Contains blisters, cracks, lumps, pinholes or fibers.	X	
	Identification marking not as specified.		X
	Surface design not as specified.		X
	Damage or defects not affecting function or serviceability.		X

4.4 Testing of the end item. The end item shall be tested for the characteristics listed in table III. The samples will be as selected in 4.2.3. The sample unit shall consist of 3 yards of matting. Each sample unit shall be selected from a different roll.

TABLE III. Test requirements

Characteristics	Reference	Test Method *
Weight of fabric	Table I	Certificate of compliance
Tensile strength	Table I	Mtd 4111 (see 4.4.1)
Resistance to sulfuric acid	Table I	Mtd 9331
Exposure to light	Table I	Mtd 5411
Air oven aging	Table I	Mtd 5111
Ultimate elongation	Table I	Mtd 4121 (see 4.4.1)
Resistance to sulfuric acid	Table I	Mtd 9331
Exposure to light	Table I	Mtd 5411
Air oven aging	Table I	Mtd 5111

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TABLE III. Test requirements (con.)

Characteristics	Reference	Test method *
Set of matting	Table I	Mtd 4311
Adhesion (classes 1 and 2)	Table I	Mtd 7211
Volatility (classes 3 and 4)	3.6.1	Mtd 9211 (see 4.4.2)
Flexibility		
At 23° ± 5° C.	3.6.2.1	Mtd 3121 (see 4.4.3)
At 0° ± 1° C.	3.6.2.2	Mtd 6511
Fire resistance	3.6.3	Mtd 6411 **
Voltage	3.6.4.1	Mtd 8111
Dielectric strength	3.6.4.2	Mtd 8211

* All test methods are found in Fed. Test Method Std. No. 501.
** Specimens will be placed on a sample holder, uncemented and the holder inserted in the horizontal flue of the machine.

4.4.1 Tensile strength and ultimate elongation. The specimen for these tests shall be dumbbell-shaped, cut by die III shown in figure 4111 of method 4111. The fabric backing or insertion shall be removed from class 1 and class 2 matting before the specimens are cut.

4.4.2 Volatility. The size of the specimen for this test shall be 4 by 4 inches in lieu of the size specified in method 9211.

4.4.3 Flexibility. When tested at 23° C. the specimen shall be 4 by 4 inches in size and the weight applied to the fold shall be 35 pounds. When testing at 0° C. the diameter of the mandrel shall be 3 ± 1/8 inch.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking of the matting shall be inspected to determine conformance to the quality assurance qualifications of specifications referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Rolls of matting furnished in lengths and widths as specified (see 6.2), shall be wrapped in paper conforming to UU-P-268, grade B, and secured with tape conforming to PPP-T-76. The rolls

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shall be overwrapped with paper conforming to PPP-B-1055, grade E2, with all folds, seams, and laps sealed with adhesive conforming to MMM-A-260. A minimum 2-inch overlap should be provided at all overlapping edges.

5.1.2 Level B (see 6.4). Rolls of matting furnished in lengths and widths as specified (see 6.2) shall be wrapped in paper conforming to UU-P-268, grade B, and secured with tape conforming to PPP-T-76. A minimum 2-inch overlap shall be provided at all overlapping edges.

5.1.3 Level C. Rolls of matting shall be wrapped in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The matting shall be packed in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-601, overseas type, or PPP-B-621, class 2.

5.2.2 Level B. The matting shall be packed in close fitting boxes conforming to PPP-B-585, class 2, PPP-B-601, domestic type, or PPP-B-621, class 1.

5.2.3 Level C. The matting shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or with National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable (see 6.2).

6. NOTES

6.1 Intended use. The matting covered by this specification is intended for use as a floor covering around electrical apparatus or circuits for protection from potential electric grounds not exceeding 3000 volts.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

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a. Title, number, and date of this specification.

b. Class required (see 1.2.1).

c. Type of wearing surface required, if different (see 3.3.3).

d. Length and width required (see 3.4.1).

e. Color required (see 3.5).

f. When certificate of compliance is not required (see 4.1.1).

g. Level of packaging and level of packing required (see 5.1 and 5.2).

h. Marking required (see 5.3).

6.3 The standard widths for this item are 24, 30, 36 and 48 inches. The length may be as long as required for installation.

6.4 Level B packaging (see 5.1.2) has been developed by General Services Administration for civil agency procurement. For Navy procurement details of level B packaging, when required, should be developed in accordance with the circumstances known at the time of procurement.

MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 84

Review activities:
Air Force - 26, 85

User activity:
Navy - MC

Preparing activity:
Navy - YD

CIVIL AGENCY INTEREST:
GSA - FSS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ZZ-M-81C - Matting, Rubber and Plastic (Special, Dielectric)

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

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